

Spatial Filter Design

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Previously...

- Mathematical operations
 - Basic, Logical operations
- Spatial transform
 - Affine
- Intensity transformation

Single
pixel

- Pixel-wise operation
 - Linear, log, gamma
- Histogram
 - Histogram Equalization
 - Histogram Match

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$s = cr^\gamma$$

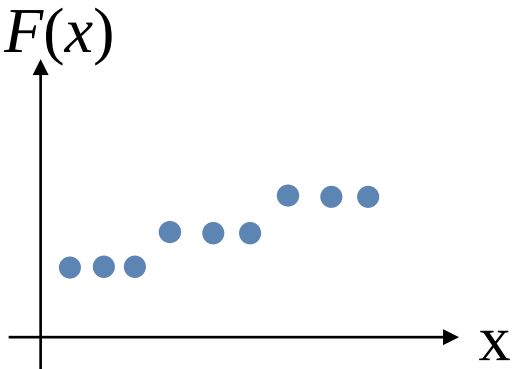
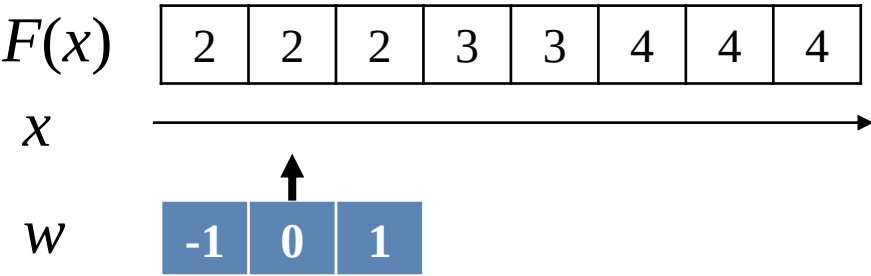
$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

Spatial filters

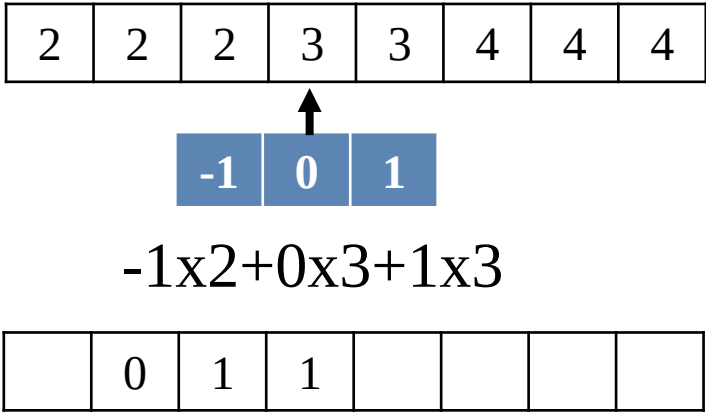
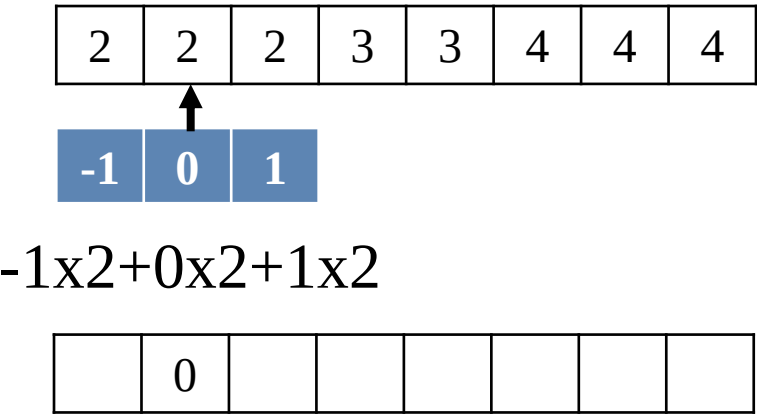
(Chapter 3.4-3.8)

- Basic of Spatial Filtering
- Spatial Filter Design
 - Low pass filter
 - High pass filter
 - Band pass filter
- Enhancement

Basic



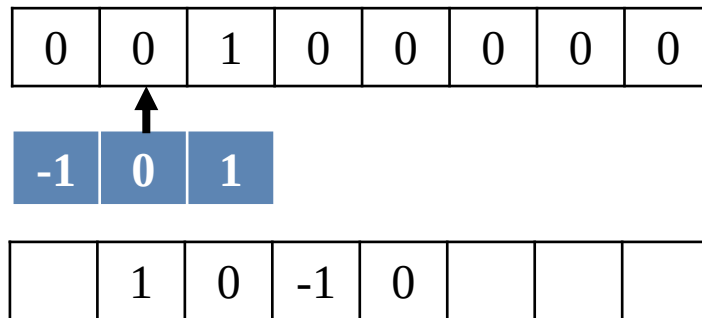
- Correlation



Basic

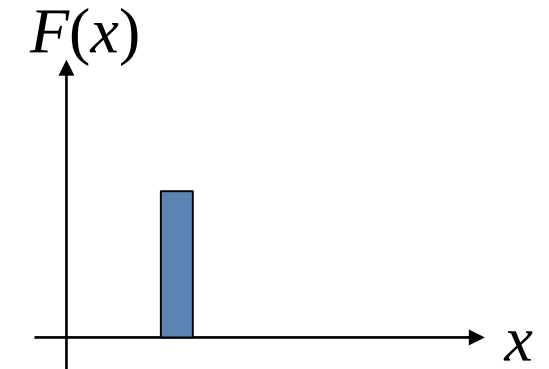
- Correlation

$$g(x) = \sum_{i=-a}^a w(i)f(i+x) \quad x \text{ is the displacement}$$

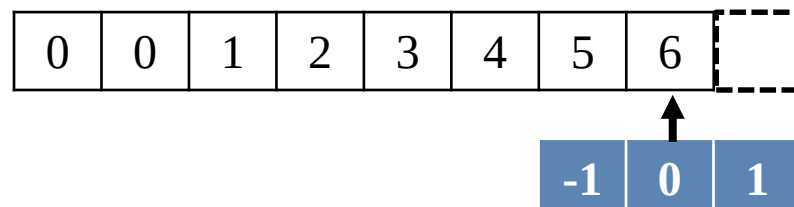


Unit Impulse

Fliped kernel



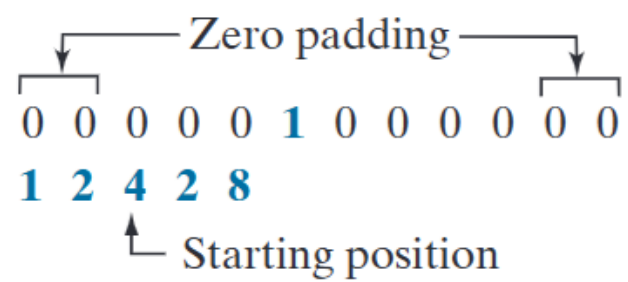
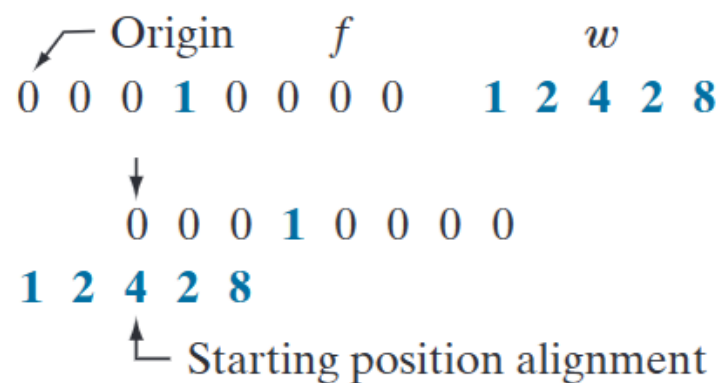
- Handling the edge



- Zero padding
- mirror

Basic

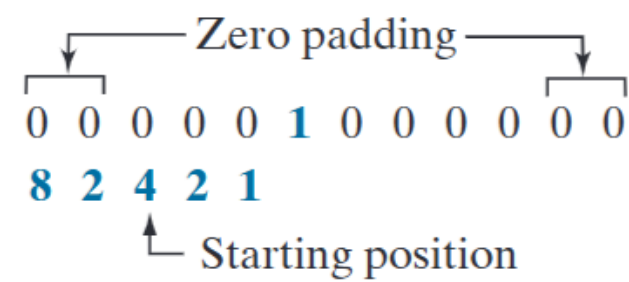
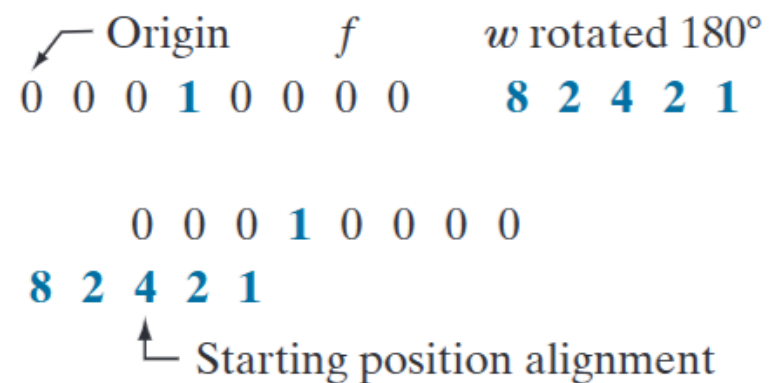
Correlation



Correlation result

0 8 2 4 2 1 0 0

Convolution



Convolution result

0 1 2 4 2 8 0 0

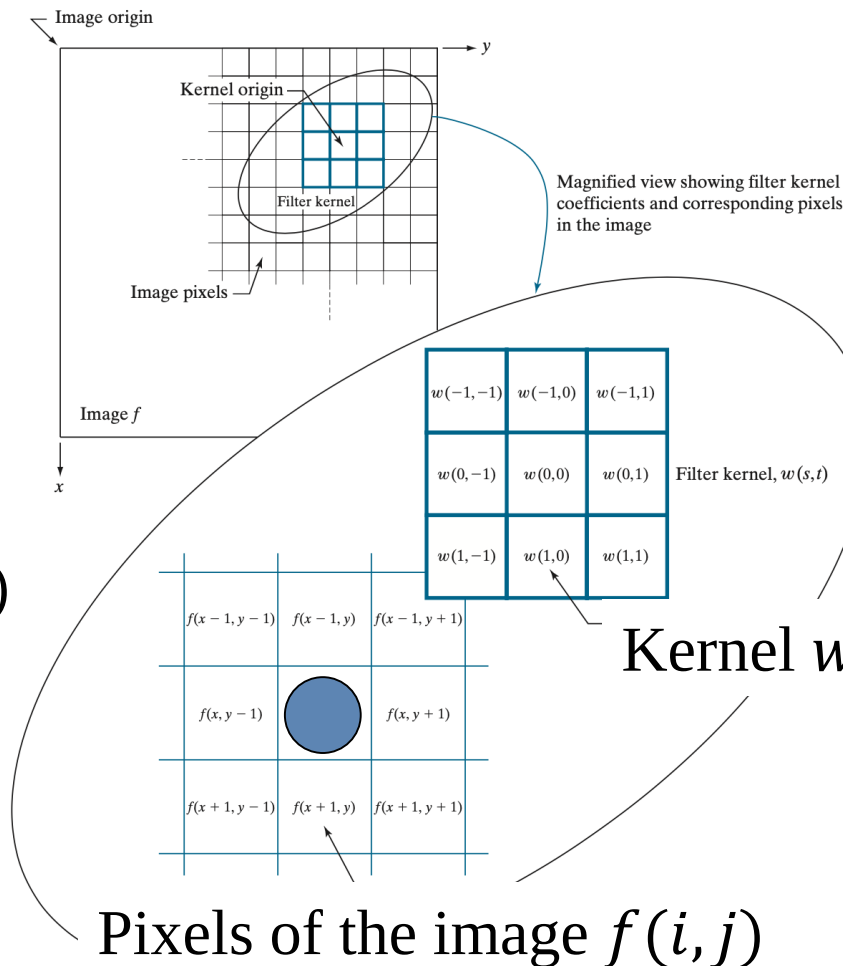
Basic

- 2D Filter: also called mask, kernel, template, window ...

- Coefficients

1	1
-1	-1

$$g(x, y) = \sum \sum w(i, j) f(i, j)$$



Size is odd number
3x3, 5x5, ...

Padded f

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

(b)

VS

(a) Origin

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

(b) Padded f

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

(c) Initial position for w

1	2	3	0	0	0	0
4	5	6	0	0	0	0
7	8	9	0	0	0	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

(d) Correlation result

0	0	0	0	0
0	9	8	7	0
0	6	5	4	0
0	3	2	1	0
0	0	0	0	0

(e) Full correlation result

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	9	8	7	0	0
0	0	6	5	4	0	0
0	0	3	2	1	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

(f) Rotated w

9	8	7	0	0	0	0
6	5	4	0	0	0	0
3	2	1	0	0	0	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

(g) Convolution result

0	0	0	0	0
0	1	2	3	0
0	4	5	6	0
0	7	8	9	0
0	0	0	0	0

(h) Full convolution result

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	1	2	3	0	0
0	0	4	5	6	0	0
0	0	7	8	9	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Basic

- Commutative: $f \star g = g \star f$
- Distributive: $f \star (g + h) = f \star g + f \star h$
- Associative: $f \star (g \star h) = (f \star g) \star h$

- Separatable

$$\mathbf{c} \mathbf{r}^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = w$$

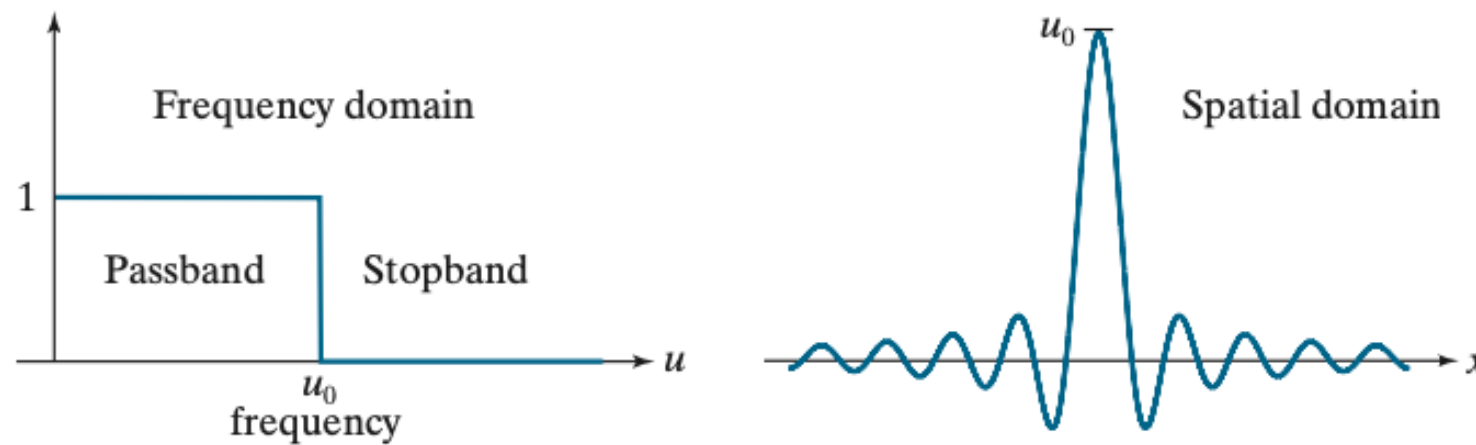
If $w = w_1 \star w_2$

$$w \star f = (w_1 \star w_2) \star f = (w_1 \star f) \star w_2$$

- Normalized

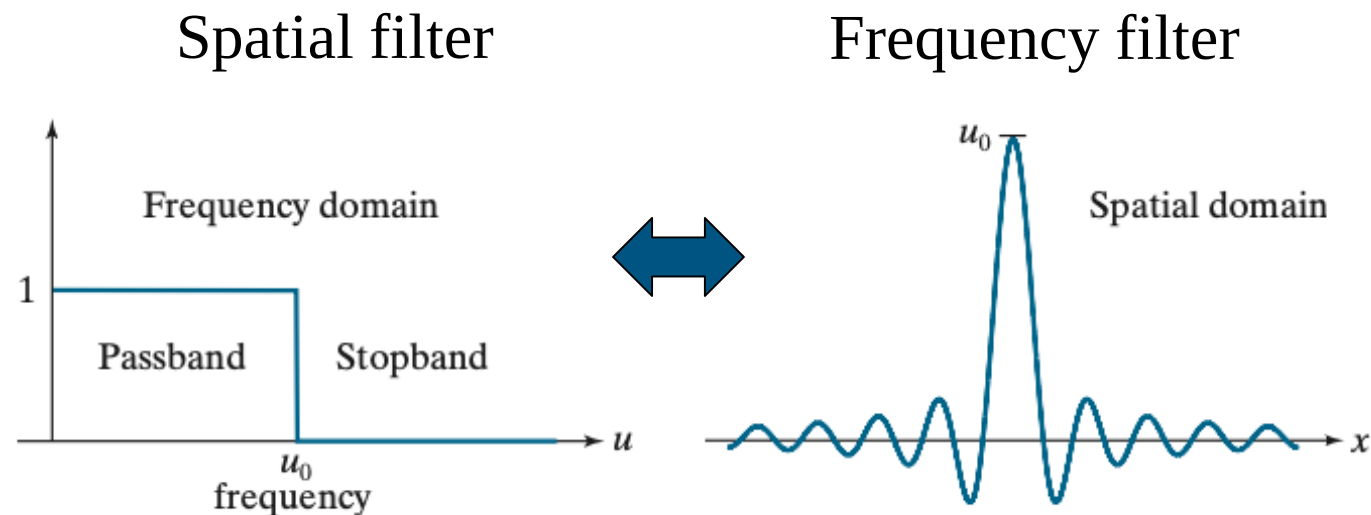
Basic

- Spatial filter and Frequency filter
 - Fourier
 - Convolution $\rightarrow$ Multiplication
 - Low/high pass



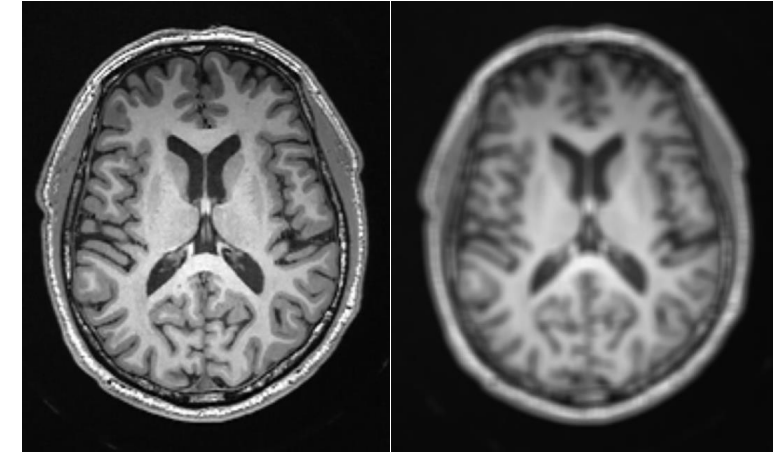
Design Approach

- Formulating filters based on mathematical properties
- Sampling a 2-D spatial function whose shape has a desired property
- Specific frequency response



Low Pass Filter

- Reduce sharp transitions in intensity
 - Reduce noise
 - Blurring
 - Reduce details before object extraction
 - Bridge small gaps
- Fundamental filter of high pass, band pass, and others



Low Pass Filter Design

- Box filter
- Gaussian filter
- Order-statistic/nonlinear filters
 - Ranking
 - Median filter (reduce impulse noise, less blurring)
 - Max filter
 - Min filter

Box Low Pass Filter

- Box filter kernel
 - Average value

$$\frac{1}{9} \times$$

1	1	1
1	1	1
1	1	1

- Weighted average

$$\frac{1}{16} \times$$

1	2	1
2	4	2
1	2	1

- Normalized

Box Low Pass Filter

Original



3x3



11x11



21x21

The dark border
resulting from zero
padding

Gaussian Low Pass Filter

- Gaussian kernel
 - Higher weights on the central pixels
 - Circularly symmetric

$$w(x, y) = G(x, y) = Ke^{\left\{-\frac{x^2+y^2}{2\sigma^2}\right\}}$$

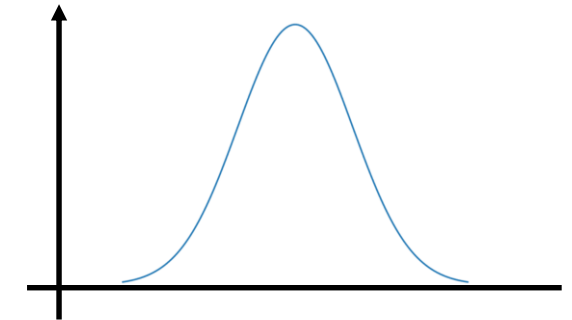
- Cropped kernel size $[-3\sigma, 3\sigma]$

- $\sigma = 7 \rightarrow 43 \times 43$

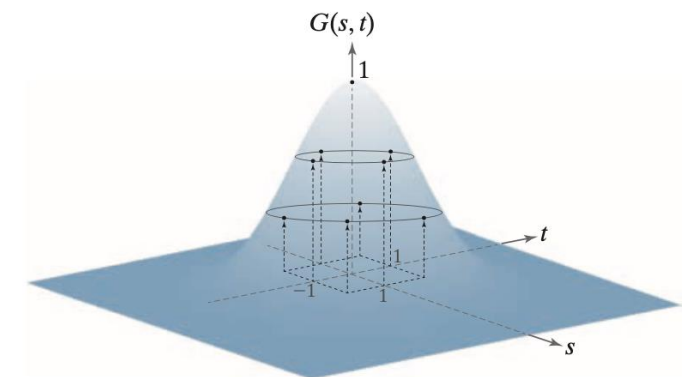
- Smooth and smooth again?

- $G1=2\text{mm}$, $G2=3\text{mm}$, $G=?$

$$\sigma_f \star g = \sqrt{\sigma_f^2 + \sigma_g^2}$$


$$\frac{1}{4.8976} \times$$

0.3679	0.6065	0.3679
0.6065	1.0000	0.6065
0.3679	0.6065	0.3679



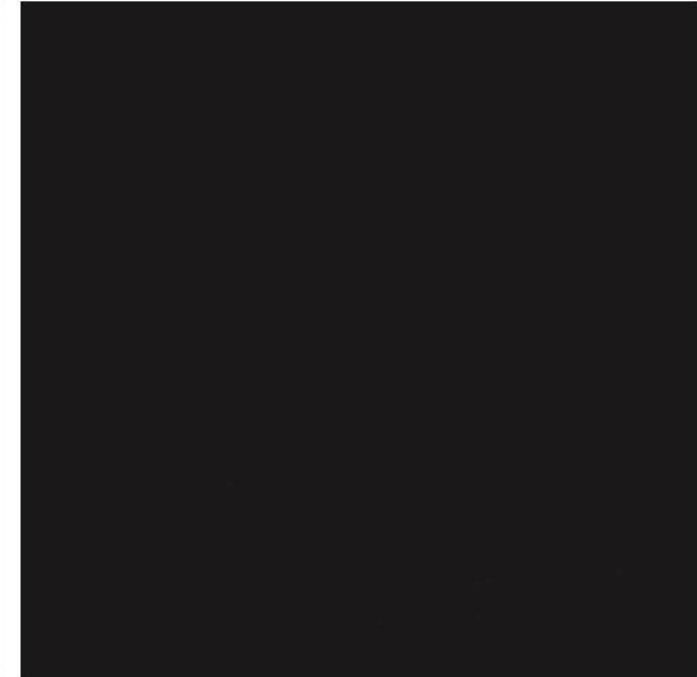
Gaussian Low Pass Filter



Gaussian
43x43
 $\sigma = 7$



Gaussian
85x85
 $\sigma = 7$



difference

Little differences beyond 3σ

Gaussian Low Pass Filter



Original



Gaussian
Kernel size=21x21
 $\sigma = 3.5$



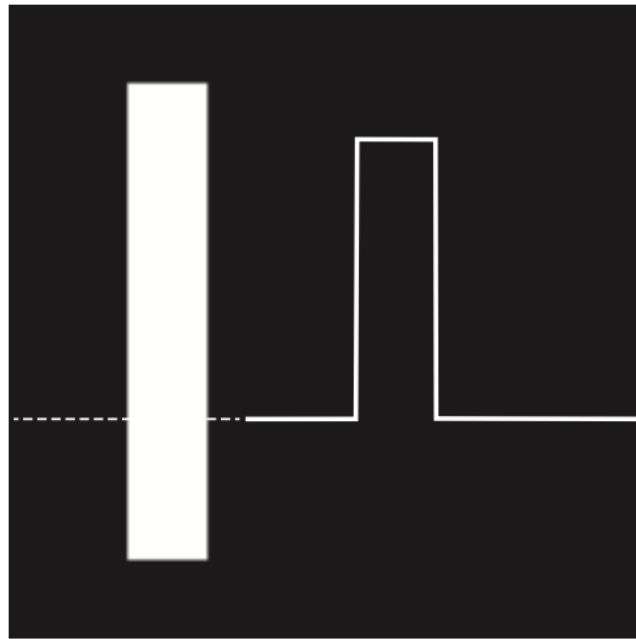
Gaussian
Kernel size=43x43
 $\sigma = 7$



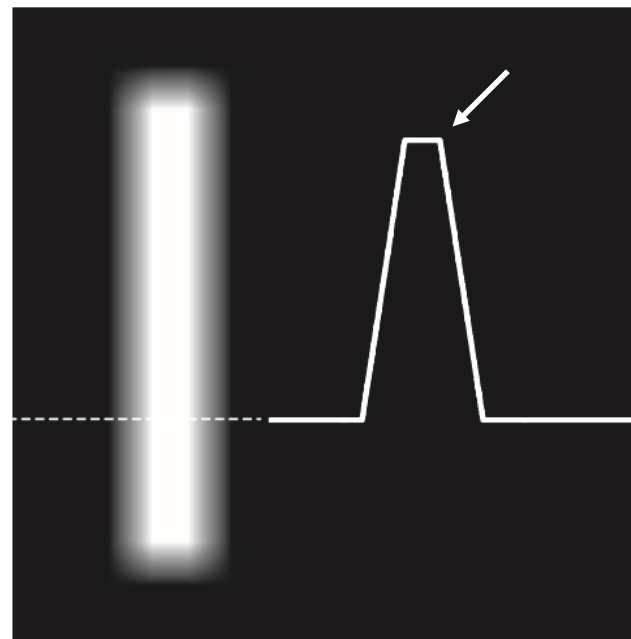
Box filter
Kernel size=21x21

Gaussian kernels have to be larger than box filters to achieve the same degree of blurring

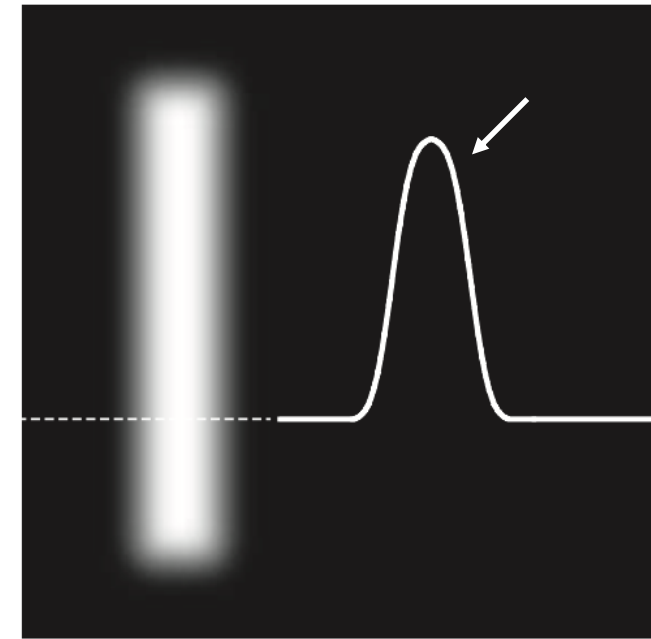
Gaussian Low Pass Filter



Original



Box
71x71



Gaussian
151x151
 $\sigma = 25$

Smooth edge transition

Low Pass Filter

- Relative size (resolution)



FOV=4096mm



Image 4096x4096

Kernel 187x187

kernel=187mm

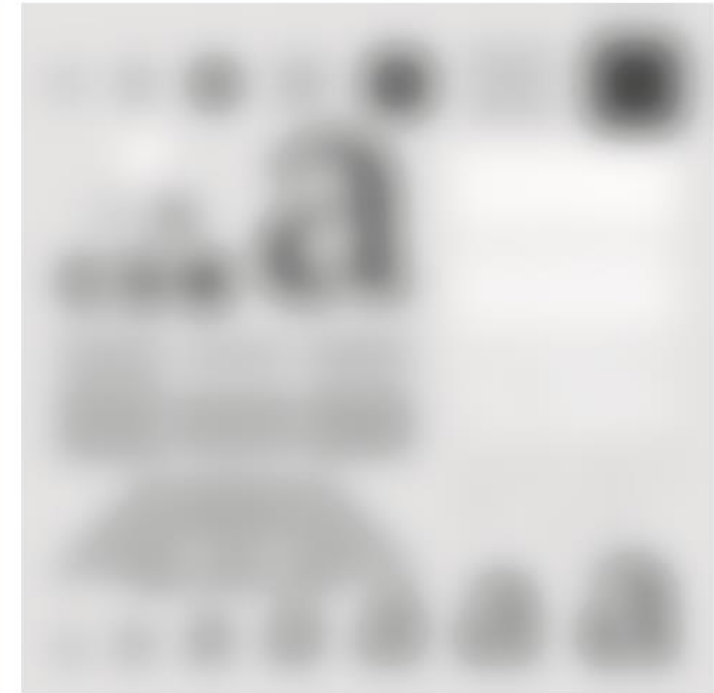


Image 1024x1024

Kernel 187x187

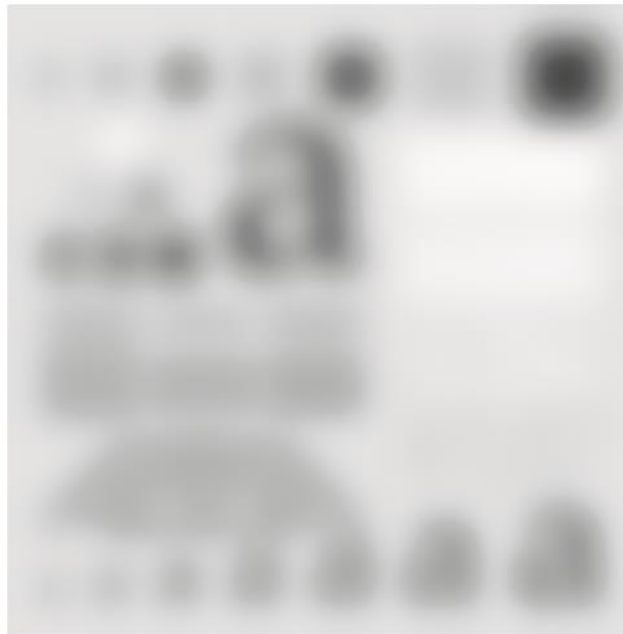
kernel = 187x4mm

Low Pass Filter

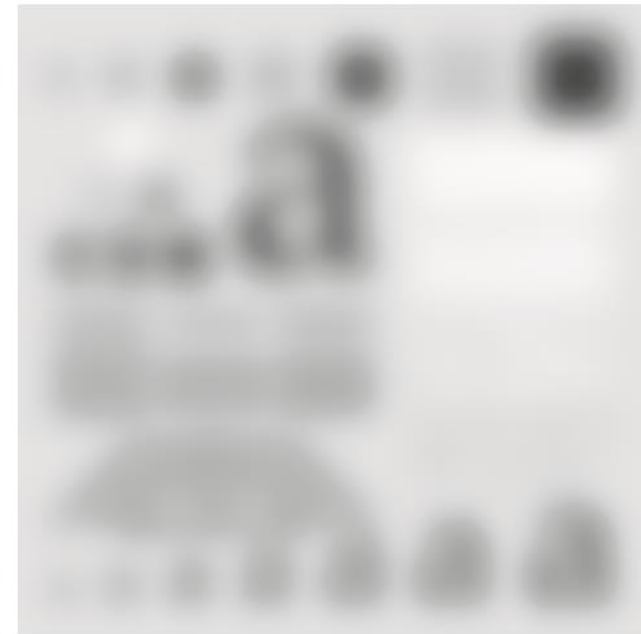
- Boundary condition



Zero-padding



Mirror



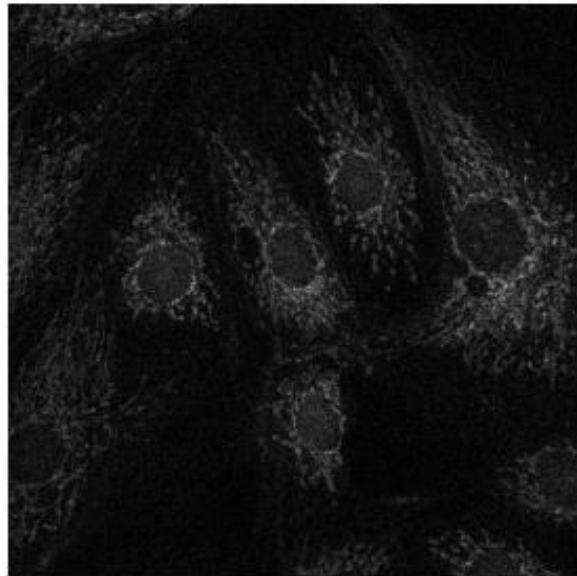
Replicated

Low Pass Filter

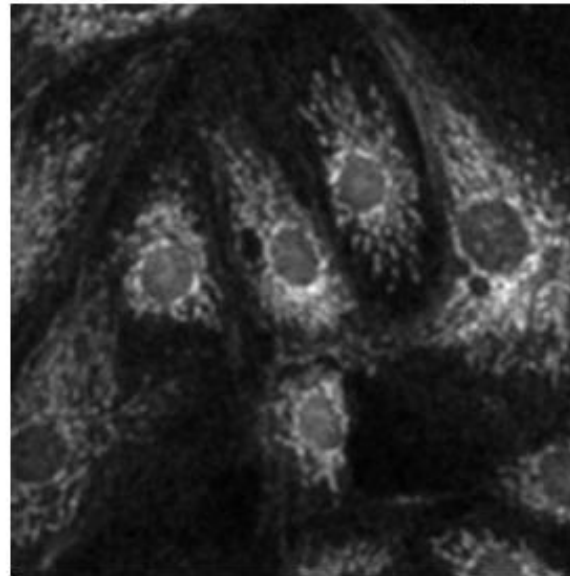
- Application: region extraction

Cell Fluorescence Microscopy Imaging

Original Image



Low-pass Filtered Image



region extraction



Low Pass Filter

- Application: shading correction

Low frequency background

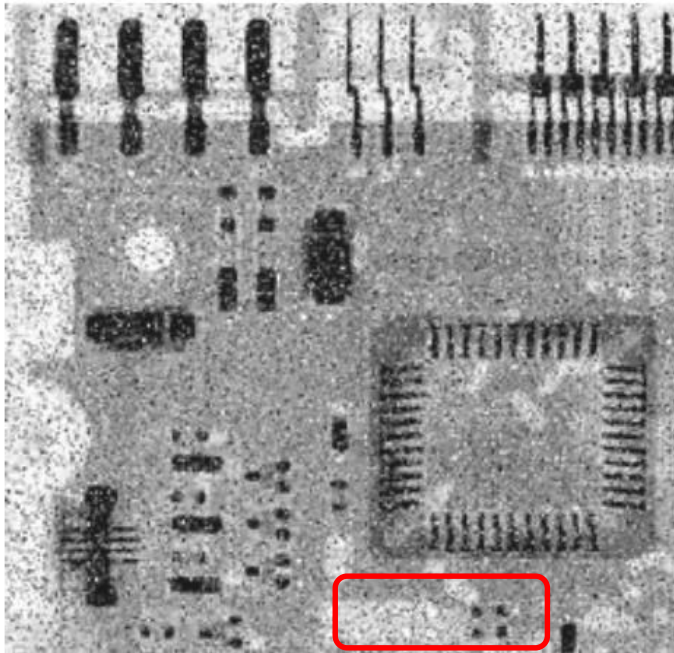


Low Pass Filter Design

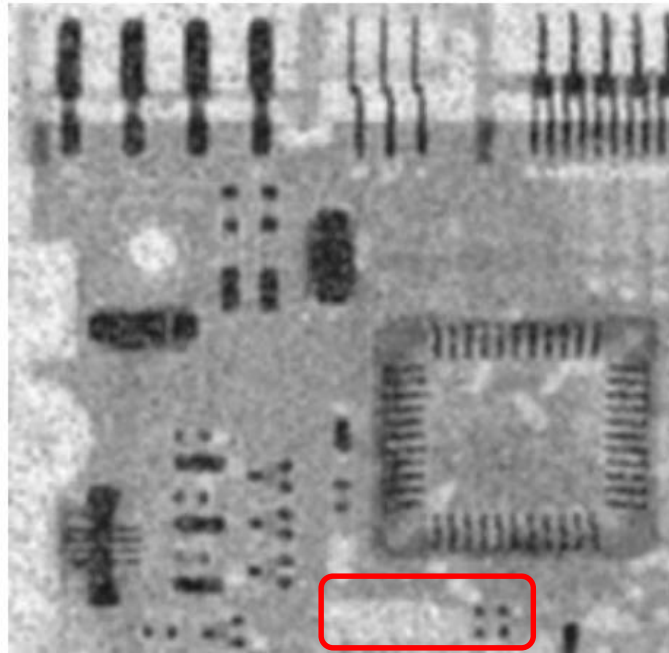
- Box filter
- Gaussian filter
- Order-statistic/nonlinear filters
 - Ranking
 - **Median filter**
 - Reduce impulse noise
 - Less blurring
 - **Max filter**
 - **Min filter**

Median Filter

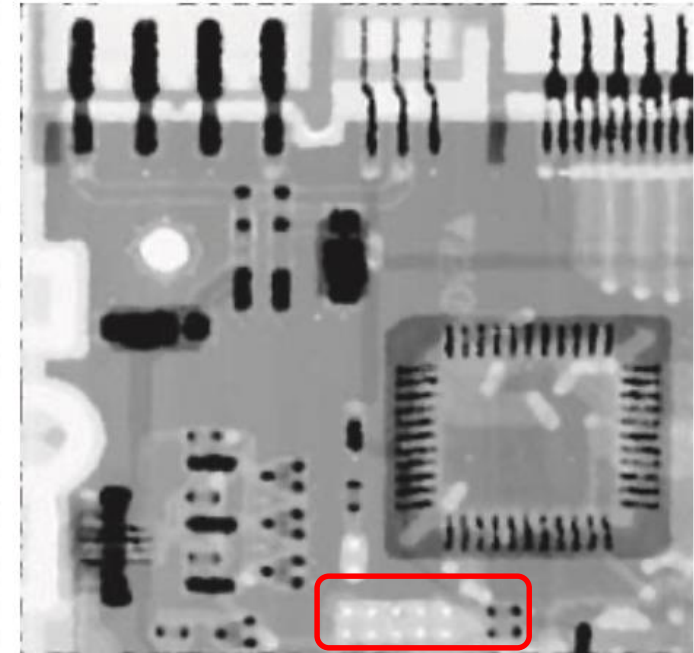
Salt-and-pepper noise



Gaussian filter



Median filter



Max and Min filter



Original Image



with Minimum Filter

2	4	3
3	2	1
4	3	5



1



Original Image



with Maximum Filter

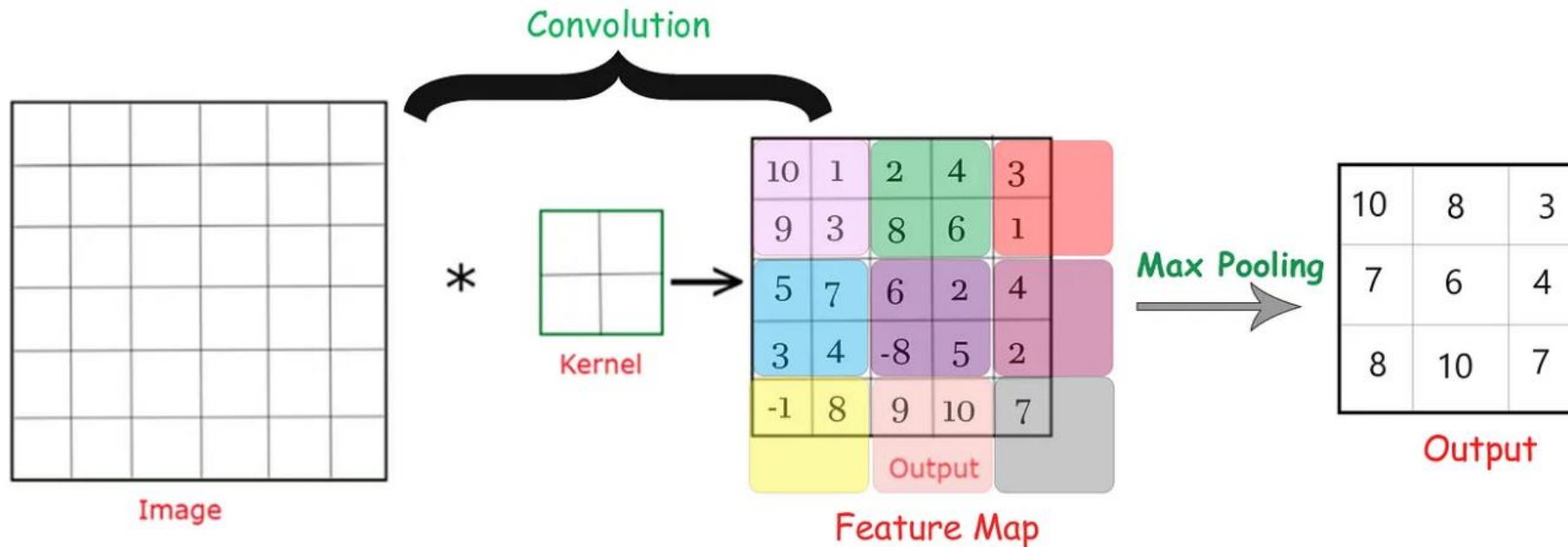
2	4	3
3	2	1
4	3	5



5

Pooling layer

- Downscaling Image by extracting most important feature
 - Stride size, Kernel size



Pooling layer

- Max pooling

(446, 450)



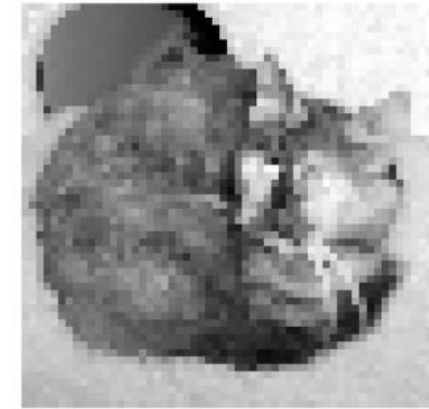
(223, 225)



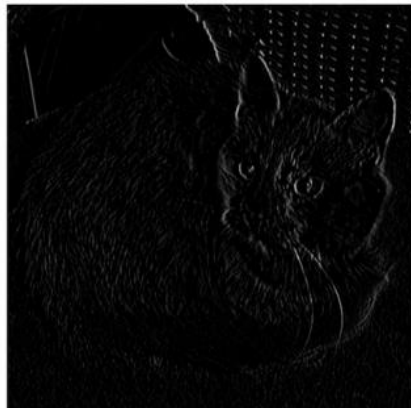
(111, 112)



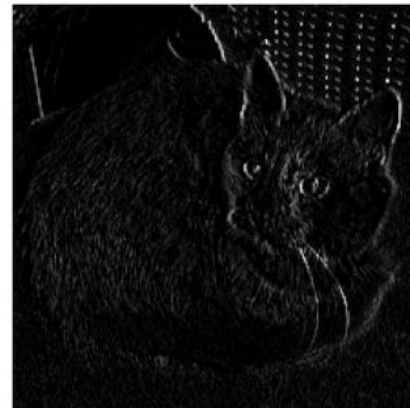
(55, 56)



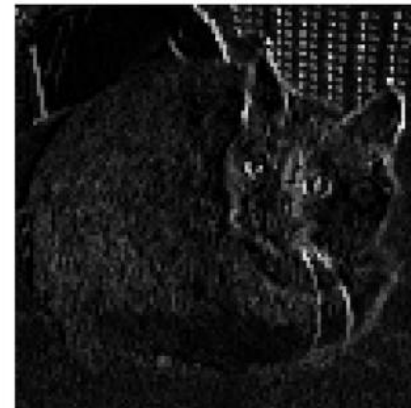
(444, 448)



(222, 224)



(111, 112)



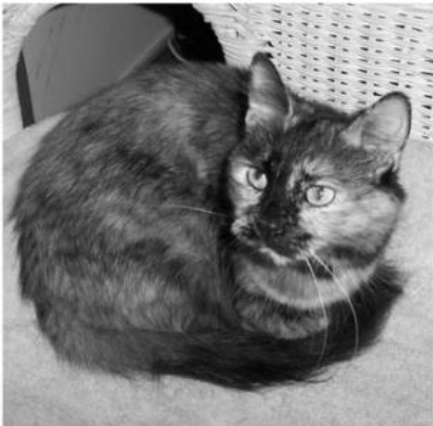
(55, 56)



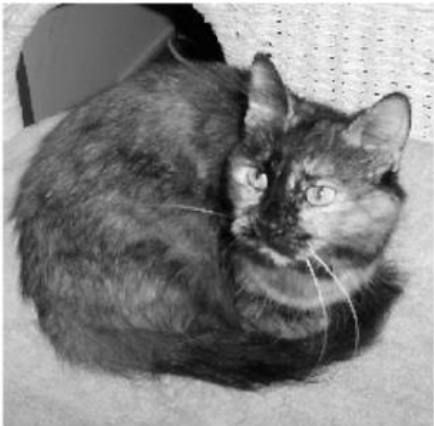
Pooling layer

Max
pooling

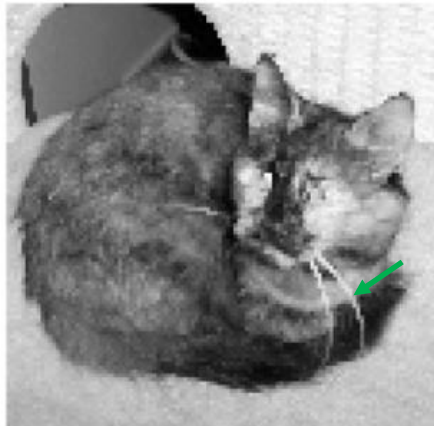
(446, 450)



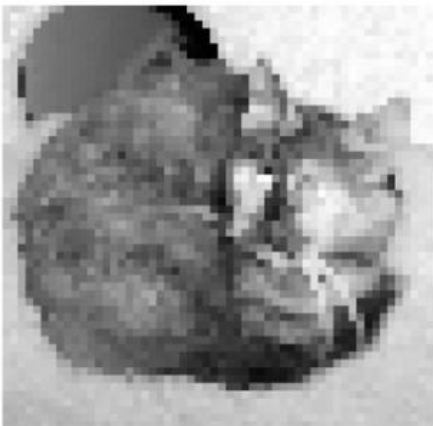
(223, 225)



(111, 112)



(55, 56)



(446, 450)



(223, 225)



(111, 112)



(55, 56)



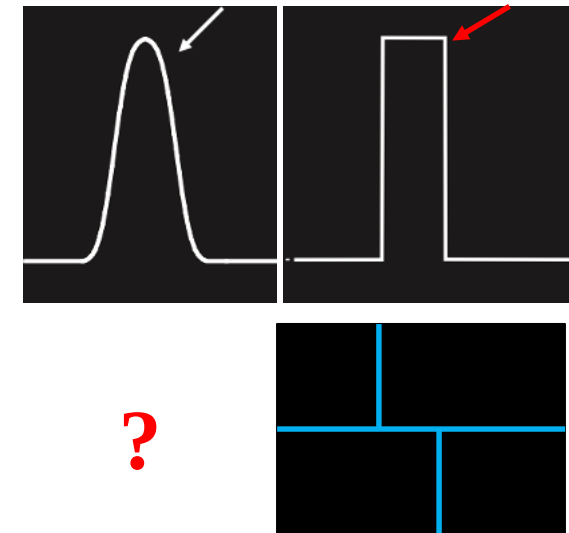
Average
pooling

Low Pass Filter Summary

- Low frequency
 - Box filter
 - Gaussian filter
 - Median filter
 - Max/Min filter
- Relative image size → resolution
- Boundary handle

High Pass Filter

- Low pass $\rightarrow$ summation / integration
- High pass $\rightarrow$ differentiation
 - Sharpening
 - Edge
 - Discontinuities
 - High frequency (fast change component)



High Pass Filter

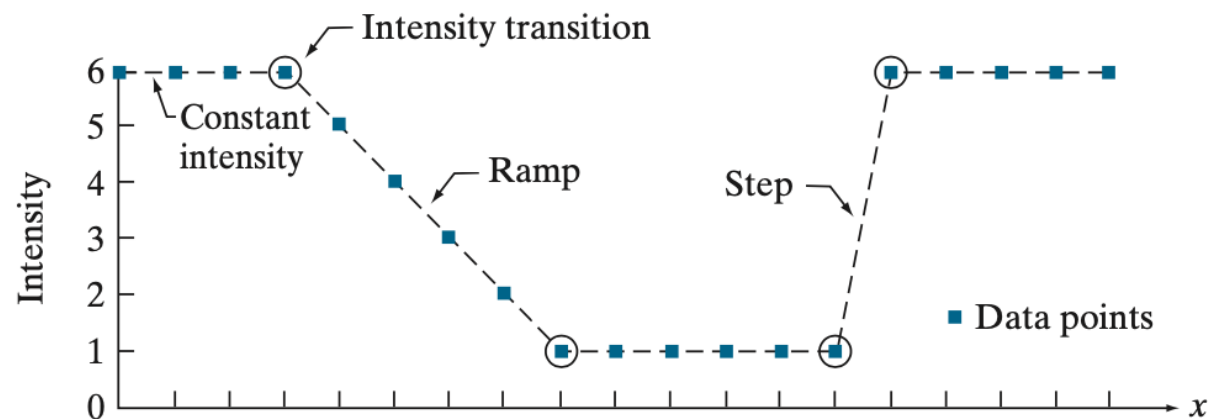
- First order

$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

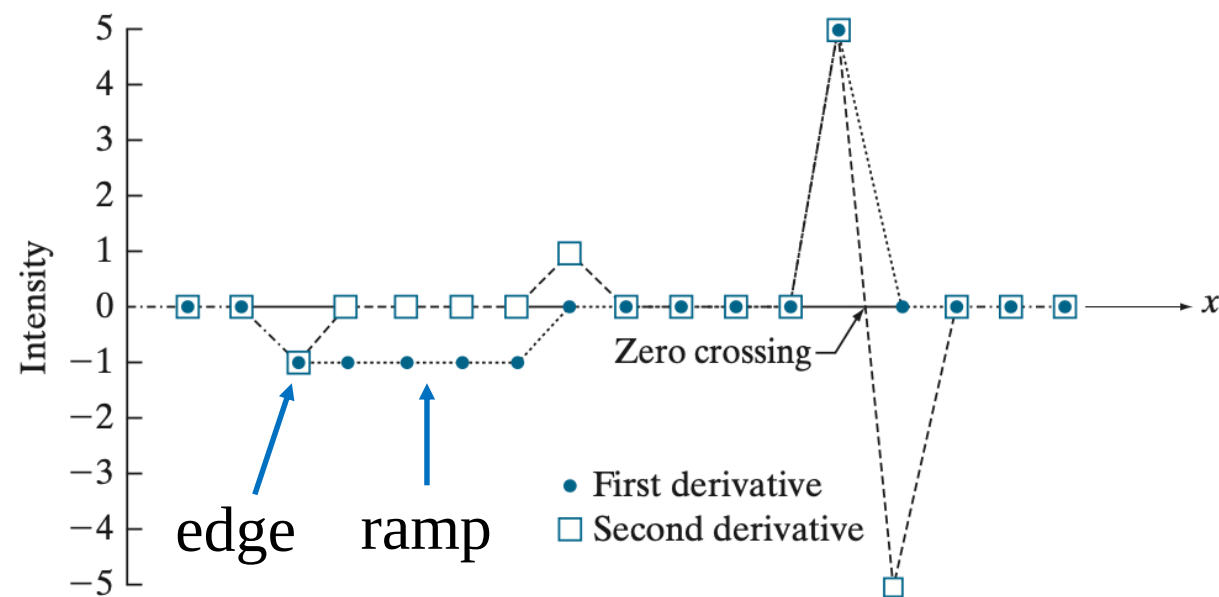
- Second order

$$\frac{\partial^2 f}{\partial x^2} = [f(x + 1) - f(x)] - [f(x) - f(x - 1)]$$

High Pass Filter



Values of scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6	x
1st derivative	0	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0	5	0	0	0	0	
2nd derivative	0	0	-1	0	0	0	0	0	1	0	0	0	0	0	5	-5	0	0	0	



High Pass Filter

- Laplacian (isotropic)

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

N4

0	1	0
1	-4	1
0	1	0

N8

1	1	1
1	-8	1
1	1	1

High Pass Filter

- Gradient (first-order derivatives)

$$\nabla f \equiv \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \quad \text{The direction of the greatest rate of change}$$

Change rate $M(x, y) = \|\nabla f\| = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$

$$M(x, y) \approx |g_x| + |g_y|$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Simplest 1st order derivative

$$G_x = z_8 - z_5$$

$$G_y = z_6 - z_5$$

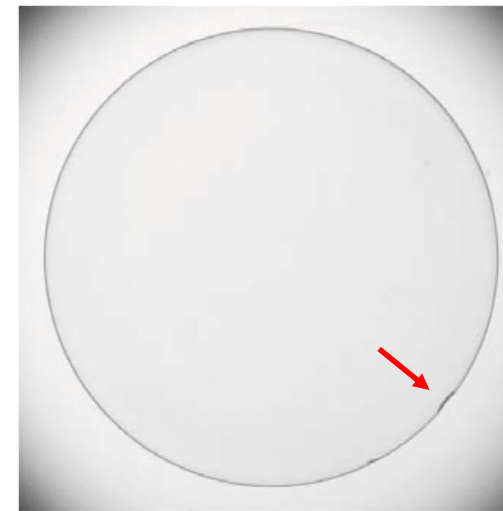
High Pass Filter

- Sobel operators

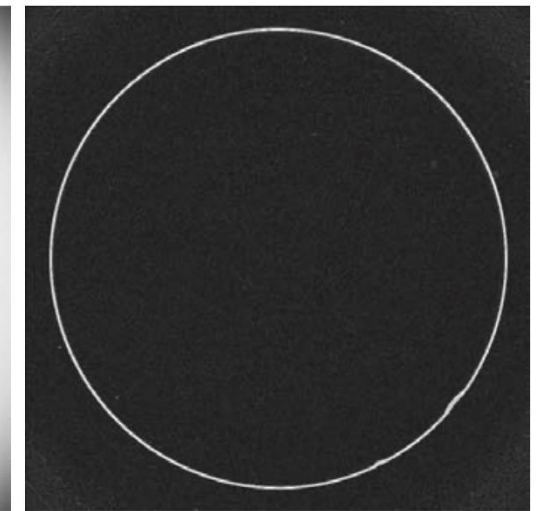
z_1	z_2	z_3	-1	-2	-1	-1	0	1
z_4	z_5	z_6	0	0	0	-2	0	2
z_7	z_8	z_9	1	2	1	-1	0	1

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$g_y = \frac{\partial f}{\partial y} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$



contact lens

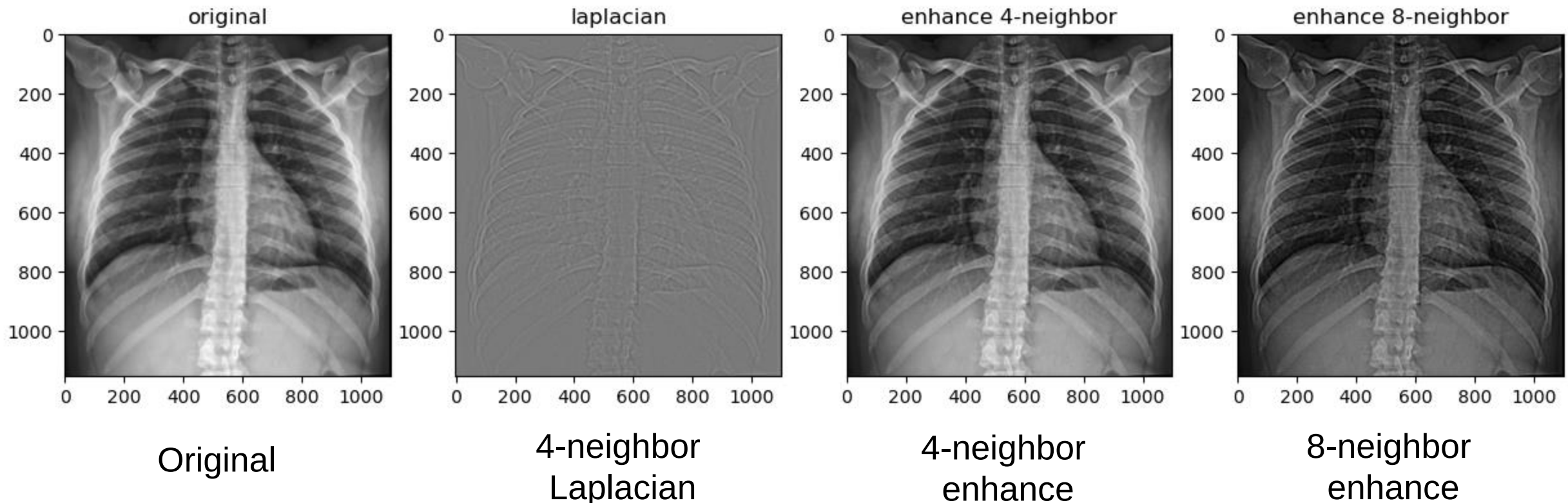


Sobel

Enhancement

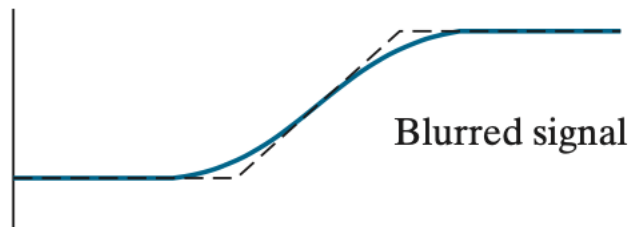
- Laplacian enhancement → highlights sharp transitions

$$g(x, y) = f(x, y) + c[\nabla^2 f(x, y)]$$



Enhancement

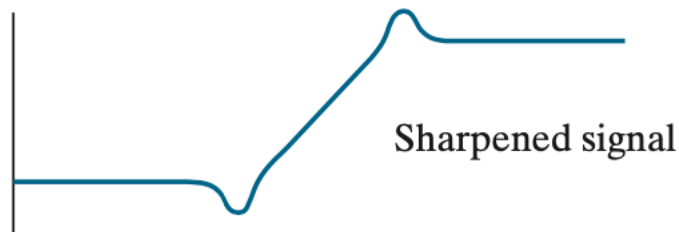
- Unsharp masking

 f  f_L 

$$f_h = f - f_L$$



$$f_{enhance} = f + kf_h$$



DIP-XE

DIP-XE

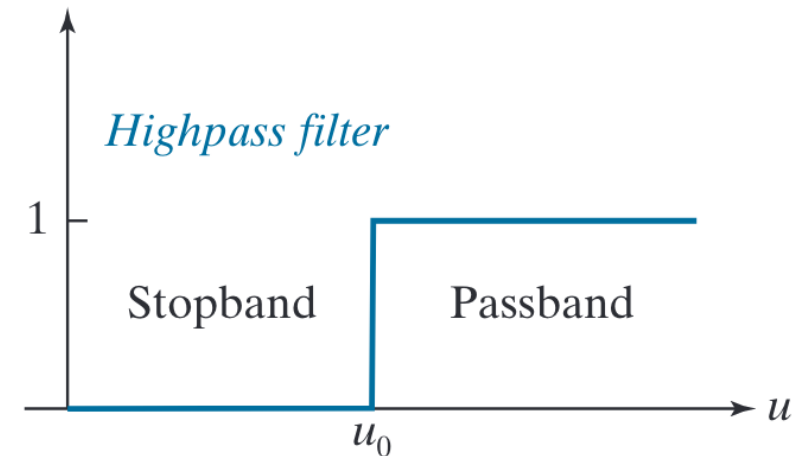
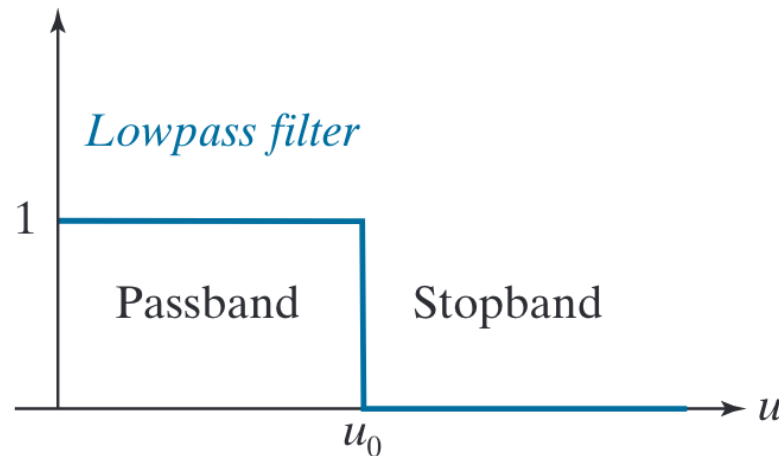
DIP-XE

DIP-XE

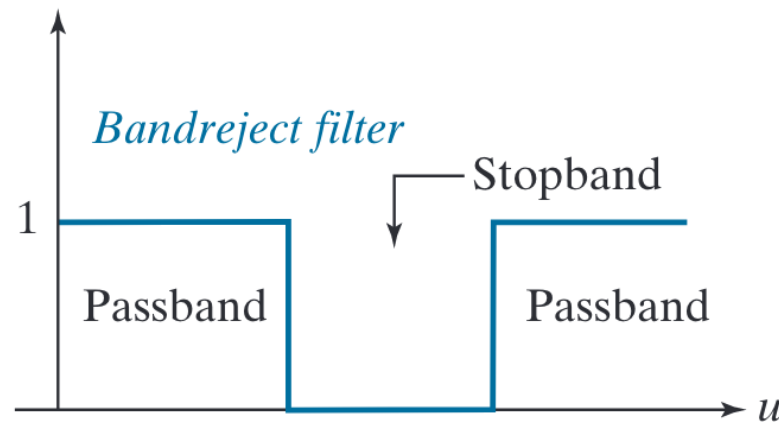
High Pass Filter Summary

- Sharping → high frequency part
 - Laplacian (second-order)
 - Gradient (first-order)
 - Sobel filter
- Enhancement
 - Laplacian enhancement
 - Unsharp mask

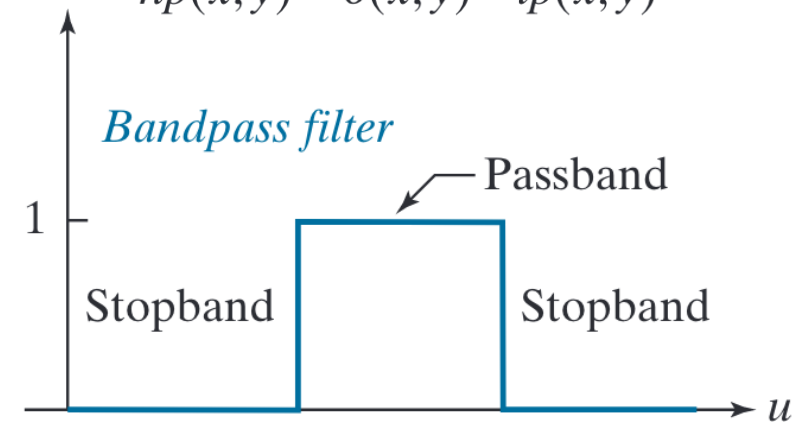
Band Pass Filter



$$hp(x, y) = \delta(x, y) - lp(x, y)$$



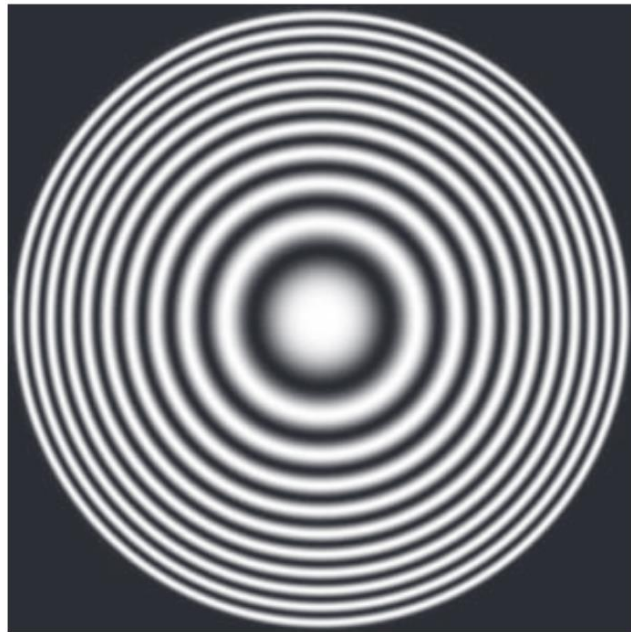
$$\begin{aligned} br(x, y) &= lp_1(x, y) + hp_2(x, y) \\ &= lp_1(x, y) + [\delta(x, y) - lp_2(x, y)] \end{aligned}$$



$$\begin{aligned} bp(x, y) &= \delta(x, y) - br(x, y) \\ &= \delta(x, y) - [lp_1(x, y) + [\delta(x, y) - lp_2(x, y)]] \end{aligned}$$

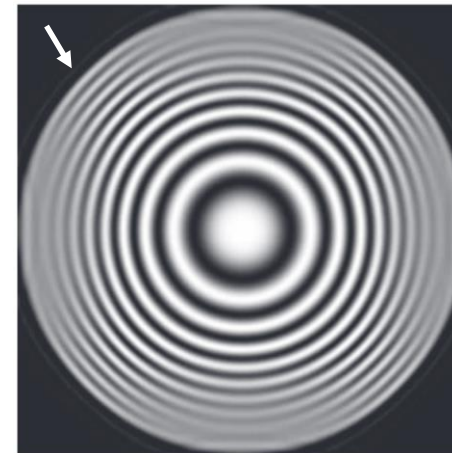
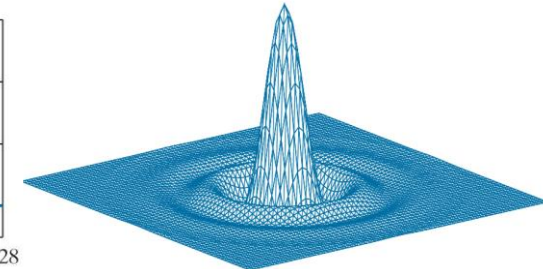
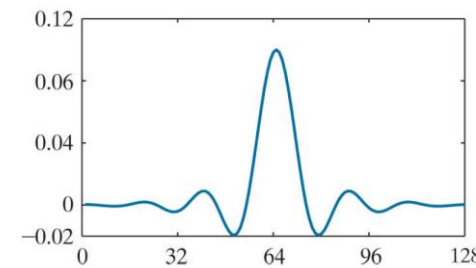
Band Pass Filter

Spatial frequency increase

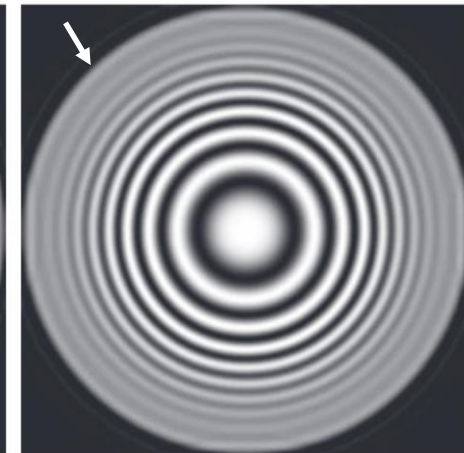


$$z(x, y) = \frac{1}{2} [1 + \cos(x^2 + y^2)]$$

Low pass filter



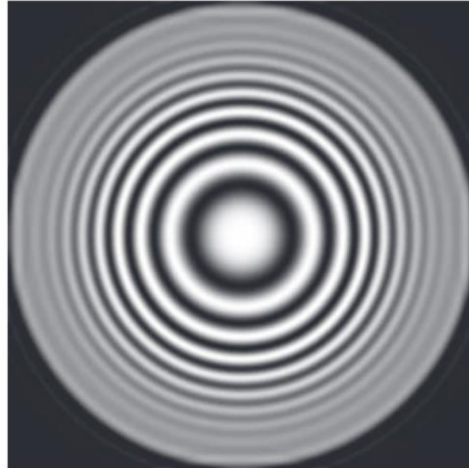
Seperatable



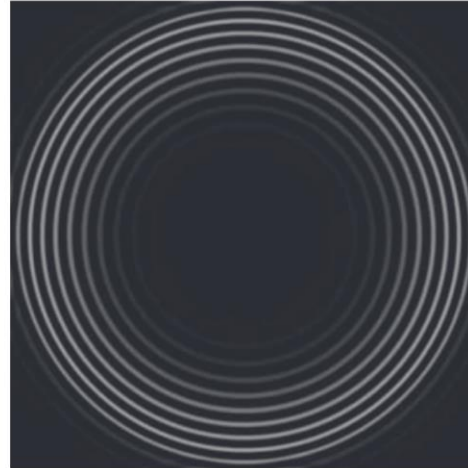
Isotropic

Band Pass Filter

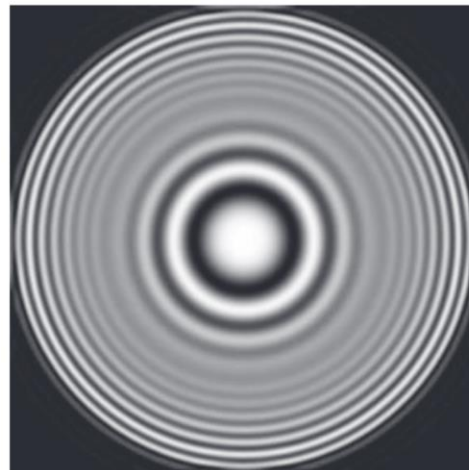
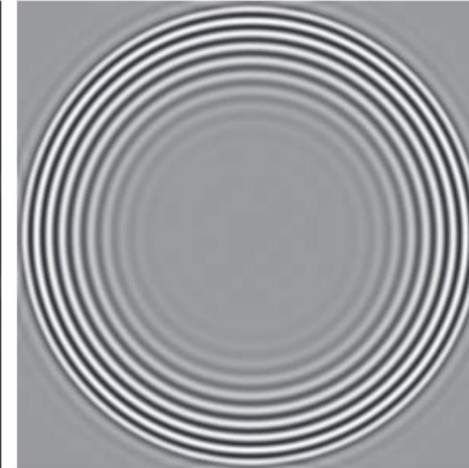
Low pass



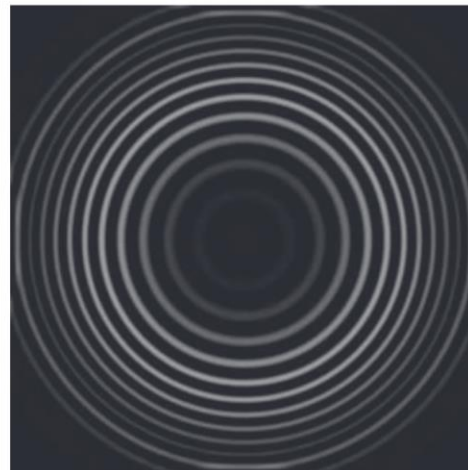
High pass



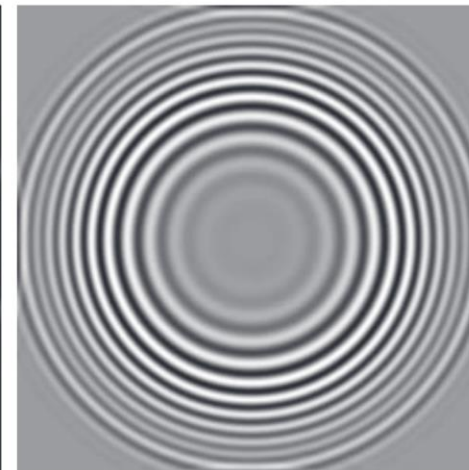
High pass (scaled)



Band reject



Band pass



Band pass (scaled)

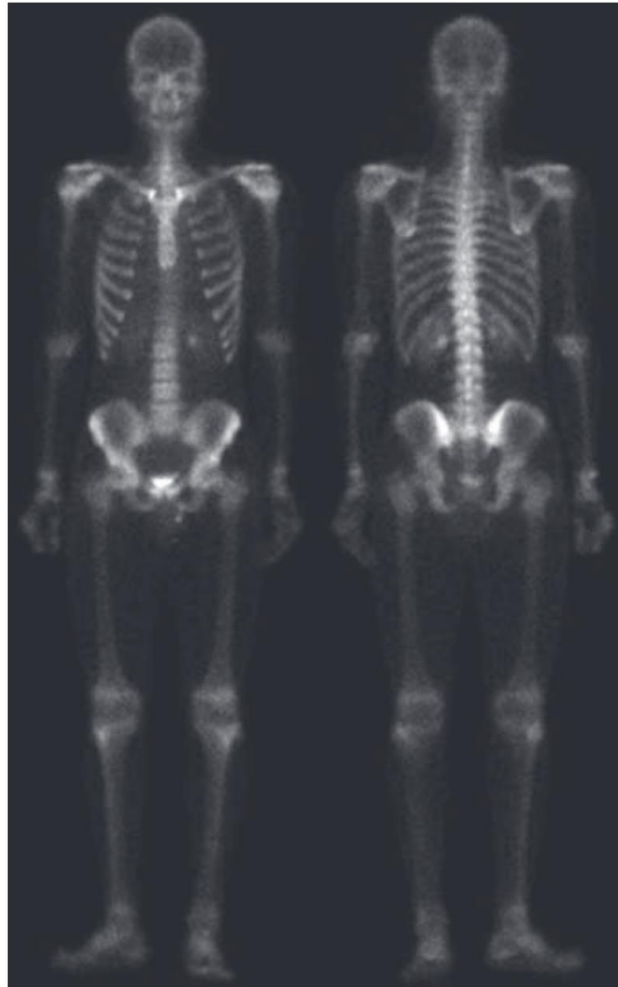
Band Pass Filter

- Frequency domain filter
- Built upon basic low pass filter

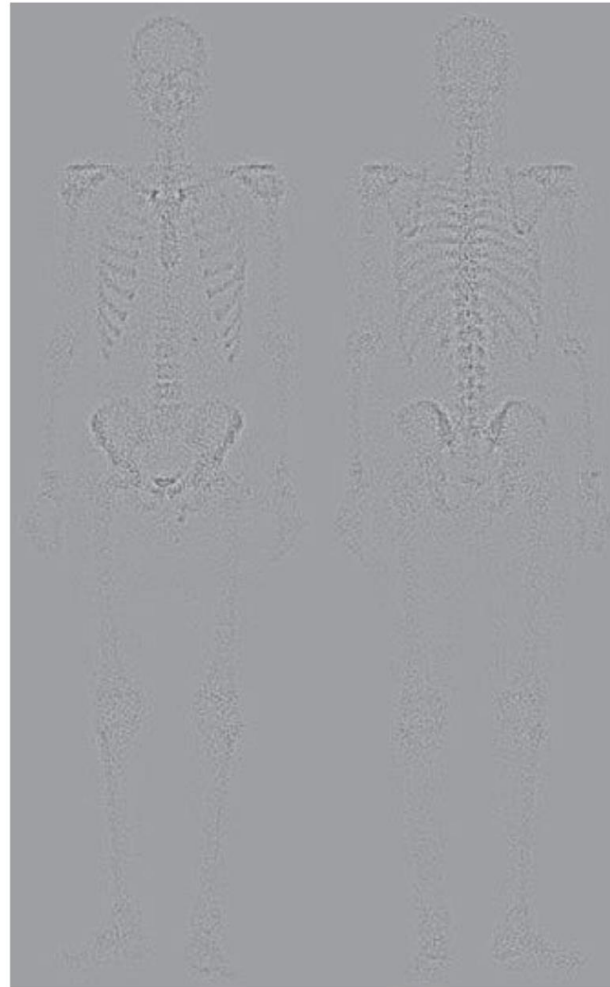
Spatial Enhancement

-1	-1	-1
-1	8	-1
-1	-1	-1

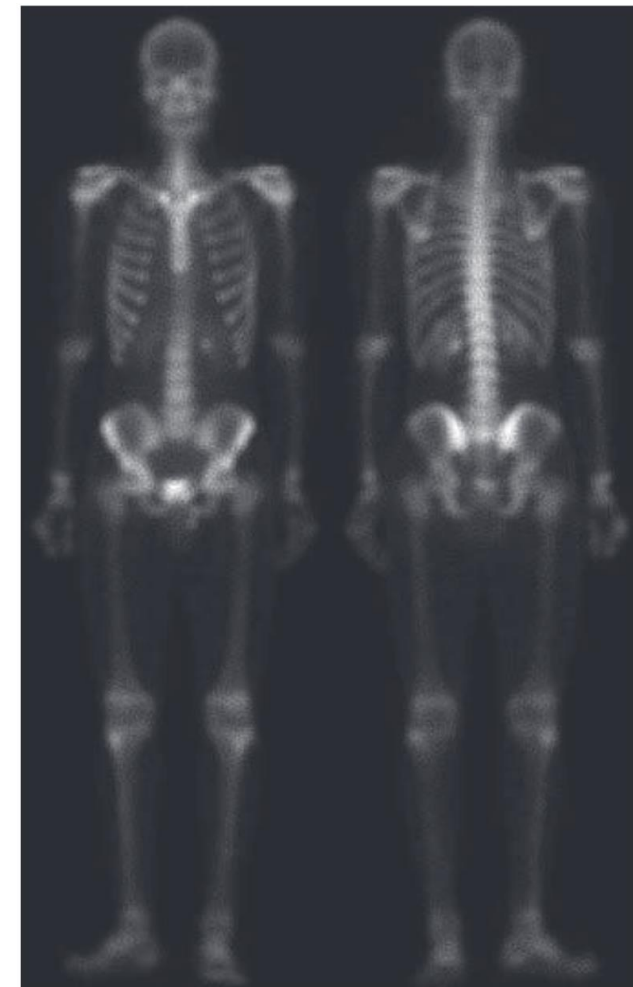
Laplacian



original



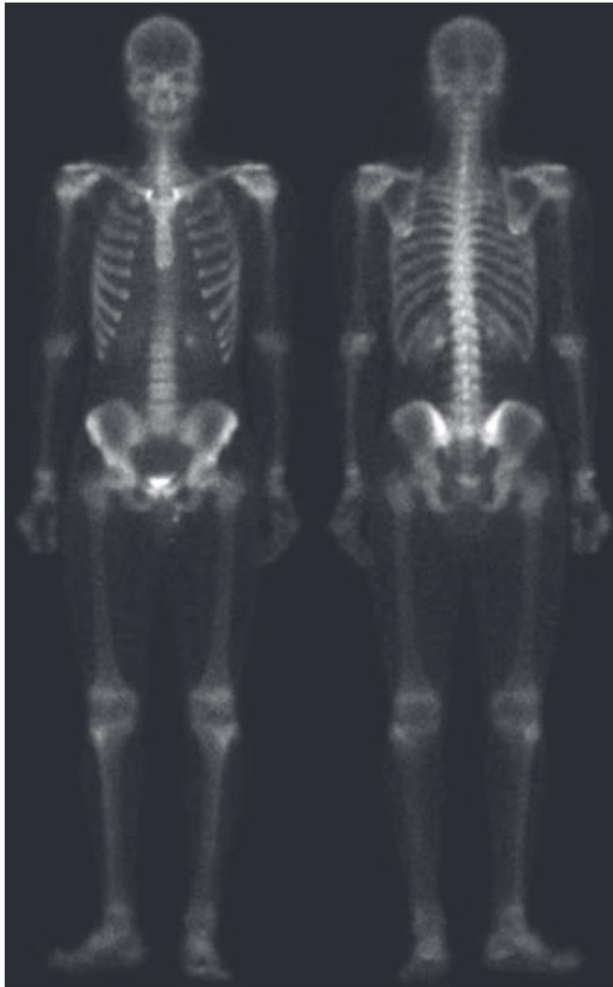
Laplacian



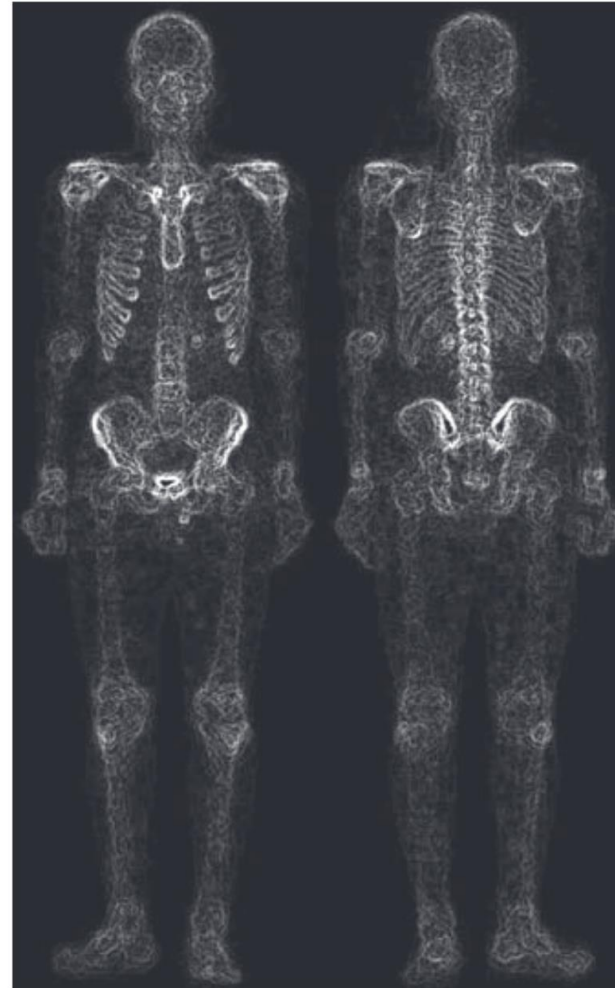
Original+Laplacian

Spatial Enhancement

original



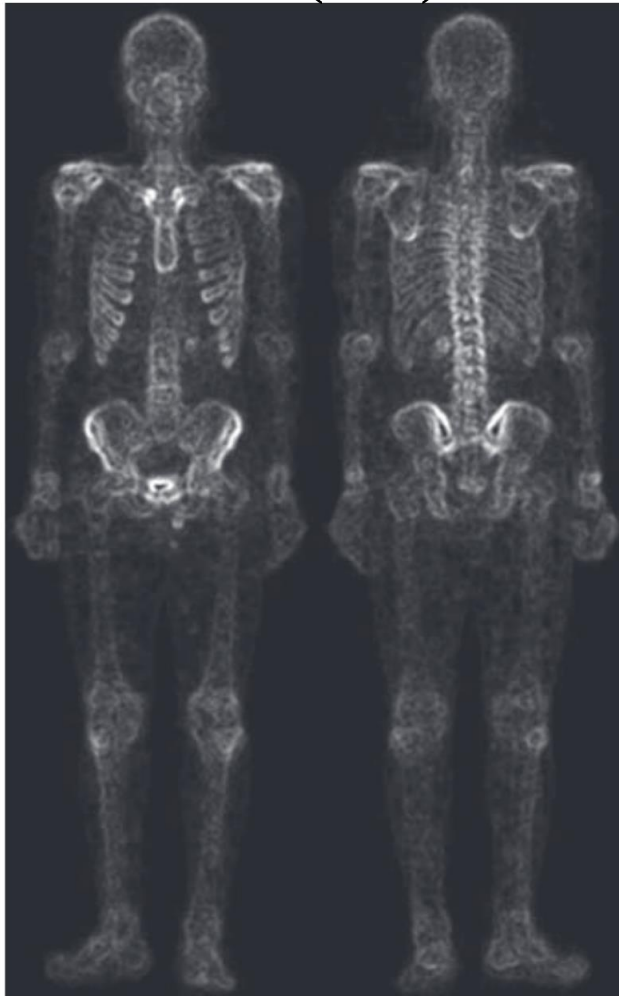
Sobel



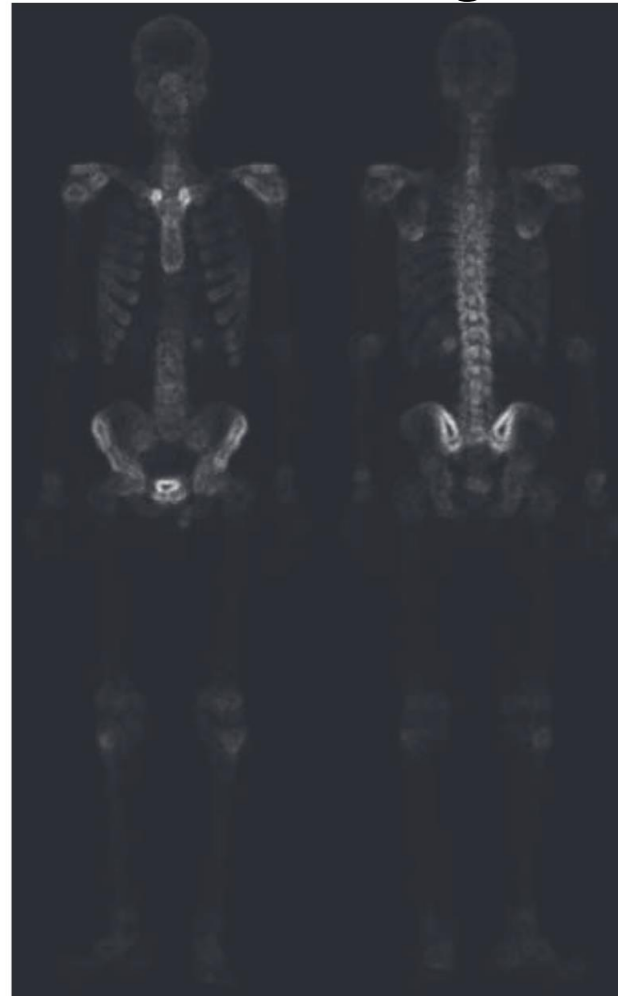
- A rather noisy sharpened image is expected.
- Median filter is incapable of removing noise in such medical images

Spatial Enhancement

Sobel (5x5)



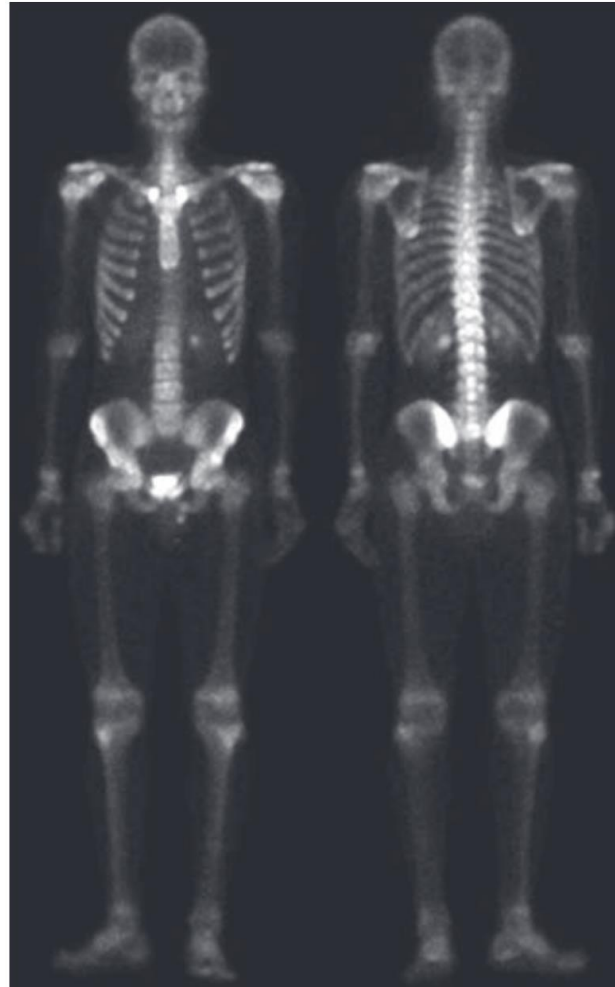
Masked image



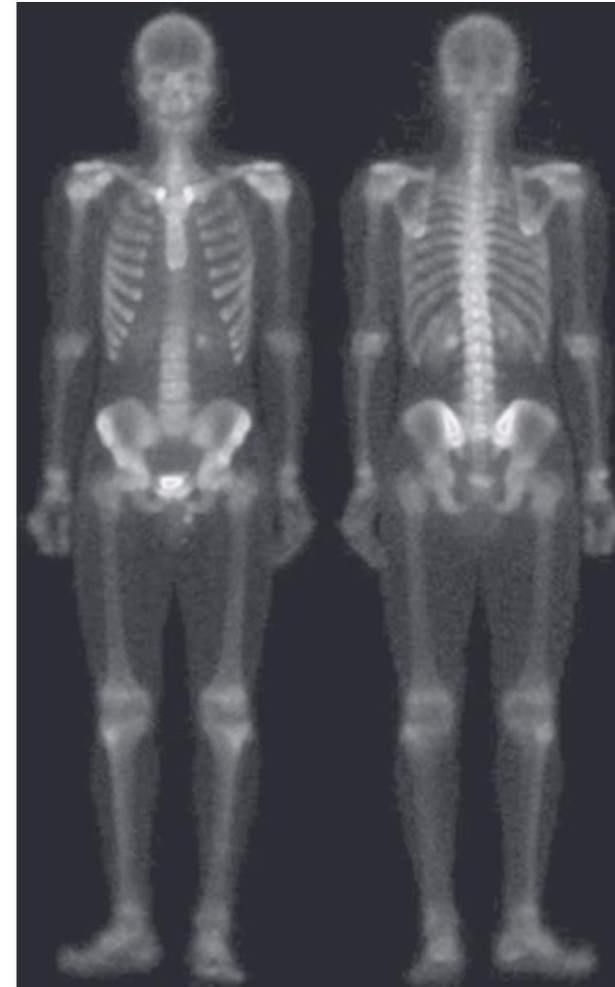
- The response of the gradient to noise and fine detail is lower than the Laplacian's and can be lowered further by smoothing the gradient with an averaging filter.
- Multiply it by the Laplacian enhanced image

Spatial Enhancement

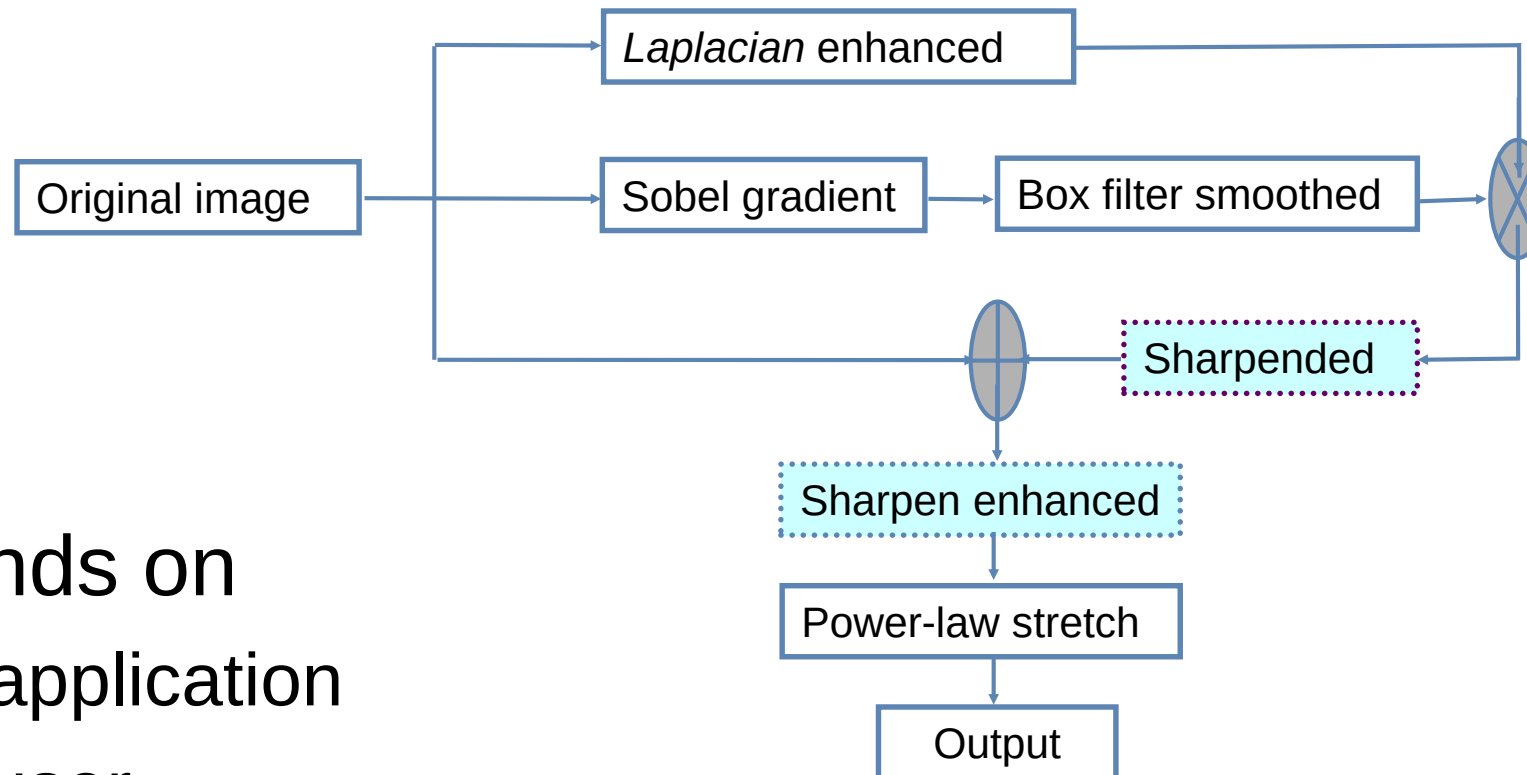
Original + Masked image



+gamma



Spatial Enhancement



- Depends on
 - the application
 - the user

Spatial filters

- Low pass filters
 - Box filter, Gaussian, Median, Max/Min
 - Denoising, smoothing
- High pass filters
 - Laplacian, Sober, Gradient
 - Enhancement
- Bandpass/Bandstop filters
- Combined spatial enhancement